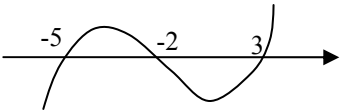


2009 NJC SH2 H2 Mathematics Prelim Paper 1 Solution

1	A graphic calculator is not to be used in answering this question. Solve the inequality $x \leq \frac{15}{x+2}$. Hence, solve the inequality $\frac{1}{x^2} \geq \frac{x^2+2}{15}$. [5]
	$x \leq \frac{15}{x+2}$ $x - \frac{15}{x+2} \leq 0$ $\frac{x^2 + 2x - 15}{x+2} \leq 0$ $\frac{(x+5)(x-3)}{x+2} \leq 0$  Ans: $x \leq -5$ or $-2 < x \leq 3$
	To solve $\frac{1}{x^2} \geq \frac{x^2+2}{15} \Rightarrow x^2 \leq \frac{15}{x^2+2}$. Replace x by x^2. Then we have $x^2 \leq -5$ (impossible) or $-2 < x^2 \leq 3$. Since $x^2 \geq 0$, we have $-\sqrt{3} \leq x \leq \sqrt{3}$, $x \neq 0$.

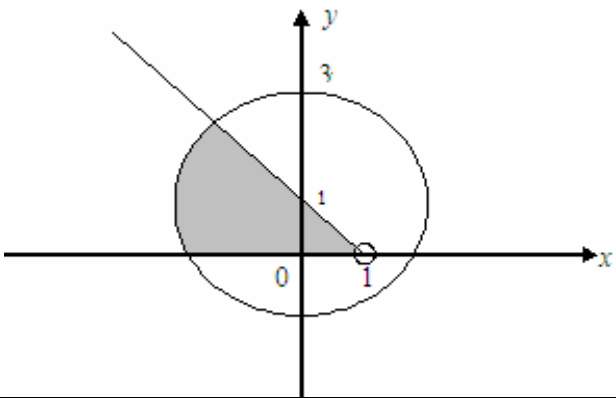
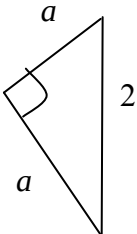
2009 NJC SH2 H2 Mathematics Prelim Paper 1 Solution

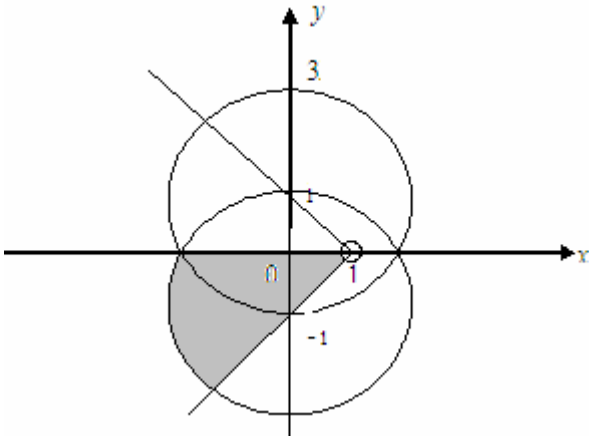
2	The function f is defined by $f : x \mapsto (2 - x)(2 + x) \quad x \in \mathbb{R}, x \geq 3.$ (i) Find the inverse of f. State clearly its domain. [3] (ii) A linear function g has domain $(-\infty, a]$. Given that gf exists such that $gf(x) = 2x^2 + 3$, find g, and state the range of possible values for a. [3]
(i)	Let $y = (2 - x)(2 + x) = 4 - x^2$ $x = \sqrt{4 - y}$ or $-\sqrt{4 - y}$ (rejected since $x \geq 3$) $f^{-1} : x \mapsto \sqrt{4 - x}$, $D_{f^{-1}} = (-\infty, -5]$
(ii)	$gf(x) = 2x^2 + 3$ $= 3 - (-2x^2 + 8 - 8)$ $= 3 - 2(4 - x^2) + 8$ $= 11 - 2(4 - x^2)$ $= 11 - 2(f(x))$ $\therefore g(x) = 11 - 2x$ For gf to exist, $R_f \subseteq D_g$ i.e. $D_g = (-\infty, -5] \therefore a \geq -5$
OR	$g(x) = gff^{-1}(x)$ $= 2(f^{-1}(x))^2 + 3$ $= 2(\sqrt{4 - x})^2 + 3$ $= 11 - 2x$
OR	By observation, $g(x) = ax + b$. $gf(x) = 2x^2 + 3$ $\Rightarrow a f(x) + b = 2x^2 + 3$ $a(4 - x^2) + b = 2x^2 + 3$ By comparing coefficient of x^2: $a = -2$ constant: $4a + b = 3 \Rightarrow b = 11$ $\therefore g(x) = 11 - 2x$

2009 NJC SH2 H2 Mathematics Prelim Paper 1 Solution

3	The points A, B and C have position vectors $3\mathbf{i} - \mathbf{j}$, $5\mathbf{i} + 2\mathbf{j} + \mathbf{k}$ and $7\mathbf{i} + 2\mathbf{j} - \mathbf{k}$ respectively. (i) Calculate the exact area of the triangle ABC. [2] The point N is on CB produced such that $BN : BC = \lambda : 1$. (ii) Find $\overrightarrow{AN}$ and determine the value of λ such that AN is perpendicular to BC. [4]
(i)	exact area of the triangle $ABC = \frac{1}{2} \overrightarrow{AB} \times \overrightarrow{AC}$ $= \frac{1}{2} \left \begin{pmatrix} 2 \\ 3 \\ 1 \end{pmatrix} \times \begin{pmatrix} 4 \\ 3 \\ -1 \end{pmatrix} \right $ $= \frac{1}{2} \left \begin{pmatrix} -6 \\ 6 \\ -6 \end{pmatrix} \right $ $= \frac{1}{2} \sqrt{36 \times 3}$ $= 3\sqrt{3}$
(ii)	$\overrightarrow{AN} = (1 + \lambda)\overrightarrow{AB} - \lambda\overrightarrow{AC}$ $= (1 + \lambda) \begin{pmatrix} 2 \\ 3 \\ 1 \end{pmatrix} - \lambda \begin{pmatrix} 4 \\ 3 \\ -1 \end{pmatrix}$ $= \begin{pmatrix} 2 \\ 3 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} -2 \\ 0 \\ 2 \end{pmatrix} = \begin{pmatrix} 2 - 2\lambda \\ 3 \\ 1 + 2\lambda \end{pmatrix}$ Since AN is perpendicular to BC, then we have $\begin{pmatrix} 2 - 2\lambda \\ 3 \\ 1 + 2\lambda \end{pmatrix} \cdot \begin{pmatrix} 2 \\ 0 \\ -2 \end{pmatrix} = 0 \Rightarrow 4 - 4\lambda - 2 - 4\lambda = 0 \Rightarrow 8\lambda = 2 \Rightarrow \lambda = \frac{1}{4}$

2009 NJC SH2 H2 Mathematics Prelim Paper 1 Solution

4	The complex number z satisfies both $iz + 1 \leq 2$ and $\frac{3\pi}{4} \leq \arg(z - 1) \leq \pi$. (i) Illustrate the locus of the points representing z in an Argand diagram. [3] (ii) Calculate the least possible value of $z - 3i$. [2] (iii) Sketch the locus of the points representing w that satisfies both $iw^* + 1 \leq 2$ and $\frac{3\pi}{4} \leq \arg(w^* - 1) \leq \pi$ on the same Argand diagram in (i). [1]
(i)	$iz + 1 \leq 2$ $i(z - i) \leq 2$ $i z - i \leq 2$ $z - i \leq 2$ 
(ii)	 $\tan\left(\frac{\pi}{4}\right) = \frac{a}{2}$ $\therefore a = \sqrt{2}$

(iii)	
5	By using the substitution $y = vx$, show that the differential equation $x \frac{dy}{dx} - y = x(x - y) \text{ where } x > 0$ reduces to $\frac{dv}{dx} = 1 - v$. [2] Find the general solution of the differential equation $x \frac{dy}{dx} - y = x(x - y)$, expressing y in terms of x. [4]
	$y = vx \Rightarrow \frac{dy}{dx} = x \frac{dv}{dx} + v$ Substituting into the given DE, we have $x \left(x \frac{dv}{dx} + v \right) - vx = x(x - vx)$ $x^2 \frac{dv}{dx} = x^2 - vx^2$ $\therefore \frac{dv}{dx} = 1 - v$ $\frac{1}{1 - v} \frac{dv}{dx} = 1$ Integrate both sides with respect to x, $\int \frac{1}{1 - v} dv = x + c$ $-\ln 1 - v = x + c$ $ 1 - v = e^{-x - c}$

2009 NJC SH2 H2 Mathematics Prelim Paper 1 Solution

	.Therefore, $1 - v = Ae^{-x}$ $\frac{y}{x} = 1 - Ae^{-x}$ $\therefore y = x - Axe^{-x}$
6	By considering $u_r - u_{r+1}$ where $u_r = \frac{1}{r(r+1)}$, show that $\sum_{r=1}^N \frac{1}{r(r+1)(r+2)} = \frac{1}{4} - \frac{1}{2(N+1)(N+2)}. \quad [3]$ (i) Deduce $\sum_{r=2}^N \frac{1}{r(r^2-1)}.$ [2] (ii) Find the value of $\frac{1}{10(10^2-1)} + \frac{1}{(11)(11^2-1)} + \frac{1}{12(12^2-1)} + \dots.$ [2]
	$u_r - u_{r+1} = \frac{1}{r(r+1)} - \frac{1}{(r+1)(r+2)} = \frac{2}{r(r+1)(r+2)}$ $\sum_{r=1}^N \frac{1}{r(r+1)(r+2)} = \frac{1}{2} \sum_{r=1}^N u_r - u_{r+1}$ $= \frac{1}{2} \left(\begin{array}{l} u_1 - u_2 \\ + u_2 - u_3 \\ + u_3 - u_4 \\ + \vdots \\ + u_{N-2} - u_{N-1} \\ + u_{N-1} - u_N \\ + u_N - u_{N+1} \end{array} \right)$ $= \frac{1}{2} (u_1 - u_{N+1})$ $= \frac{1}{2} \left(\frac{1}{2} - \frac{1}{(N+1)(N+2)} \right)$ $= \frac{1}{4} - \frac{1}{2(N+1)(N+2)}$

2009 NJC SH2 H2 Mathematics Prelim Paper 1 Solution

(i)	Replace r by $R + 1$, $\sum_{r=2}^N \frac{1}{r(r^2 - 1)} = \sum_{r=2}^N \frac{1}{r(r+1)(r-1)}$ $= \sum_{R=1}^{N-1} \frac{1}{R(R+1)(R+2)}$ $= \frac{1}{4} - \frac{1}{2N(N+1)}$
(ii)	$\frac{1}{10(10^2 - 1)} + \frac{1}{(11)(11^2 - 1)} + \frac{1}{12(12^2 - 1)} + \dots$ $= \sum_{r=10}^{\infty} \frac{1}{r(r^2 - 1)}$ $= \sum_{r=2}^{\infty} \frac{1}{r(r^2 - 1)} - \sum_{r=2}^9 \frac{1}{r(r^2 - 1)}$ $= \frac{1}{4} - \left(\frac{1}{4} - \frac{1}{2(9)(10)} \right)$ $= \frac{1}{180}$

2009 NJC SH2 H2 Mathematics Prelim Paper 1 Solution

7 (a) (b)	In a triangle ABC, $AB = 3$ cm and $AC = 2$ cm. If angle BAC is increasing at a constant rate at 0.1 radians per second, find the rate of increase of the length BC at the instant where $BAC = \frac{\pi}{3}$ radians. [4] Suppose p is a positive constant. Find the coordinates of the points on the curve $x^2 - xy = p^2 + y^2$ at which the normal is parallel to the x-axis. Leave your answers in terms of p. [4]
(a)	Let $BC = x$ cm and $\angle BAC = \theta$ radians. By cosine rule, $x^2 = 3^2 + 2^2 - 2(3)(2)\cos\theta$ $x^2 = 13 - 12\cos\theta$ $2x \frac{dx}{dt} = 12 \sin\theta \frac{d\theta}{dt}$ When $\theta = \frac{\pi}{3}$ and $\frac{d\theta}{dt} = 0.1$, $x = \sqrt{7}$ $2\sqrt{7} \frac{dx}{dt} = 12 \left(\frac{\sqrt{3}}{2} \right) (0.1)$ $\frac{dx}{dt} = \frac{0.3\sqrt{3}}{\sqrt{7}} \approx 0.196 \text{ cm/s}$
(b)	$x^2 - xy = p^2 + y^2$ $2x - x \frac{dy}{dx} - y - 2y \frac{dy}{dx} = 0$ $\frac{dy}{dx} = \frac{2x - y}{2y + x}$ If the normal is parallel to the x-axis, then its tangent is parallel to the y-axis. That is $2y + x = 0$ $x = -2y$ Substitute back to the original equation: $(-2y)^2 + 2y^2 = p^2 + y^2$ $5y^2 = p^2$ $y = \pm \frac{p}{\sqrt{5}}$

2009 NJC SH2 H2 Mathematics Prelim Paper 1 Solution

	Then $x = \mp \frac{2p}{\sqrt{5}}$. Thus the coordinates are $(-\frac{2p}{\sqrt{5}}, \frac{p}{\sqrt{5}})$ and $(\frac{2p}{\sqrt{5}}, -\frac{p}{\sqrt{5}})$
8	(a) Write down constants A and B such that, for all values of x, $x = A(2x + 4) + B$. Find $\int \frac{x}{x^2 + 4x + 6} dx$. [5] (b) (i) Show that $\frac{d}{dx} \left(\frac{1}{\sqrt{x^2 - 1}} \right) = -\frac{x}{(x^2 - 1)^{\frac{3}{2}}}$. [1] (ii) By using the substitution $x = \frac{1}{y}$, find $\int \frac{1}{x\sqrt{x^2 - 1}} dx$. [3] Find $\int x(x^2 - 1)^{-\frac{3}{2}} \ln x dx$. [2]
(a)	$x = \frac{1}{2}(2x + 4) - 2$. Hence $A = \frac{1}{2}$ and $B = -2$.
	$\int \frac{x}{x^2 + 4x + 6} dx$ $= \frac{1}{2} \int \frac{2x + 4}{x^2 + 4x + 6} dx - 2 \int \frac{1}{(\sqrt{2})^2 + (x + 2)^2} dx$ $= \frac{1}{2} \ln(x^2 + 4x + 6) - \frac{2}{\sqrt{2}} \tan^{-1} \left(\frac{x + 2}{\sqrt{2}} \right) + c$ $= \ln \sqrt{x^2 + 4x + 6} - \sqrt{2} \tan^{-1} \left(\frac{x + 2}{\sqrt{2}} \right) + c$

(b) (i)	$\frac{d}{dx} \left(\frac{1}{\sqrt{x^2 - 1}} \right) = -\frac{1}{2} (2x) (x^2 - 1)^{-\frac{3}{2}}$ $= -\frac{x}{(x^2 - 1)^{\frac{3}{2}}}$
	$x = \frac{1}{y} \Rightarrow \frac{dx}{dy} = -\frac{1}{y^2}$ $\int \frac{1}{x\sqrt{x^2 - 1}} dx = \int \frac{1}{\left(\frac{1}{y}\right) \sqrt{\left(\frac{1}{y}\right)^2 - 1}} \left(-\frac{1}{y^2}\right) dy$ $= \int -\frac{1}{y} \div \sqrt{\frac{1 - y^2}{y^2}} dy$ $= \int \frac{-1}{\sqrt{1 - y^2}} dy$ $= -\sin^{-1}(y)$ $= -\sin^{-1}\left(\frac{1}{x}\right) + C$ OR $\cos^{-1}\left(\frac{1}{x}\right) + C$
	$\int x(x^2 - 1)^{-\frac{3}{2}} \ln x \, dx.$ $= \left(\frac{-1}{\sqrt{x^2 - 1}} \right) \ln x - \int \frac{-1}{\sqrt{x^2 - 1}} \left(\frac{1}{x} \right) dx$ $= \left(\frac{-1}{\sqrt{x^2 - 1}} \right) \ln x + \int \frac{1}{x\sqrt{x^2 - 1}} dx$ $= \left(\frac{-1}{\sqrt{x^2 - 1}} \right) \ln x - \sin^{-1} \left(\frac{1}{x} \right) + C$ OR $\left(\frac{-1}{\sqrt{x^2 - 1}} \right) \ln x + \cos^{-1} \left(\frac{1}{x} \right) + C$

2009 NJC SH2 H2 Mathematics Prelim Paper 1 Solution

9	(a) Given that $z = -\sqrt{3} + i$, express z in the form $re^{i\theta}$ where r is real and $-\pi < \theta \leq \pi$. Hence find exactly the values of a and b which satisfy the equation $(iz)^4 = e^{a+ib}$, where $a \in \mathbf{R}$, and $-\pi < b \leq \pi$. [4] (b) Solve $w^3 + 8 = 0$ where w is a complex number. Leave your answers in the exact form $x + iy$ where x and y are real and $-\pi < \theta \leq \pi$. [3] Deduce the roots of $w^4 + w^3 + 8w + 8 = 0$ [2] Hence, determine the roots of z such that $z^4 - iz^3 + 8iz + 8 = 0$. [2]
(a)	$ z = 2 \quad \arg(z) = \pi - \frac{\pi}{6} = \frac{5\pi}{6}$ $\therefore z = 2e^{i\frac{5\pi}{6}}$ $(iz)^4 = e^{a+ib}$ $i^4 z^4 = e^a e^{ib}$ $2^4 e^{i\left(\frac{5\pi}{6}\right)^4} = e^a e^{ib}$ $e^a = 2^4 \qquad e^{ib} = e^{i\left(\frac{5\pi}{6}\right)^4}$ $\therefore a = 4 \ln 2 \qquad \therefore b = \frac{20\pi}{6} \equiv -\frac{2\pi}{3}$
(b)	$w^3 = -8$ $w^3 = 8e^{i(2k\pi+\pi)}$ $\Rightarrow w = 2e^{i\left(\frac{2k\pi+\pi}{3}\right)}$ $\therefore k = 0, w = 2e^{i\left(\frac{\pi}{3}\right)} = 2\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right) = 1 + i\sqrt{3}$ $k = 1, w = 2e^{i\pi} = 2(\cos\pi + i\sin\pi) = -2$ $k = -1, w = 2e^{i\left(-\frac{\pi}{3}\right)} = 2\left(\cos\left(-\frac{\pi}{3}\right) + i\sin\left(-\frac{\pi}{3}\right)\right) = 1 - i\sqrt{3}$ $\therefore w = -2, 1 + i\sqrt{3}, 1 - i\sqrt{3}$
	$w^4 + w^3 + 8w + 8 = 0$ $(w+1)(w^3 + 8) = 0$ $\therefore w = -1, -2, 1 + i\sqrt{3}, 1 - i\sqrt{3}$

2009 NJC SH2 H2 Mathematics Prelim Paper 1 Solution

	$z^4 - iz^3 + 8iz + 8 = 0$ Let $w = iz$, $\therefore z = i, 2i, \sqrt{3} - i, -\sqrt{3} - i$
10	The points A and B have position vectors $-\mathbf{i} + 2\mathbf{j} - 2\mathbf{k}$ and $2\mathbf{i} - \mathbf{j} + 7\mathbf{k}$ respectively. The plane Π_1 is given by $\mathbf{r} \cdot (\mathbf{i} - \mathbf{j}) = 5$. (i) Another plane Π_2 is perpendicular to Π_1 and contains the points A and B. Find the equation of Π_2 in the form $\mathbf{r} \cdot \mathbf{n} = d$. [3] (ii) Find $\overline{ON}$, where N is the foot of perpendicular from the point A to the plane Π_1. [3] (iii) Deduce the equation of the line of intersection between the planes Π_1 and Π_2. [2] The plane Π_3 has vector equation $\mathbf{r} \cdot (4\mathbf{i} + 2\mathbf{j} - \mathbf{k}) = D$, where D is a positive constant. (iv) Determine if the planes Π_1, Π_2 and Π_3 meet at common point or a common line. [2] (v) Given that the shortest distance of the point B from the plane Π_3 is $\sqrt{21}$, find the value of D. [3]
(i)	$\overline{AB} = \begin{pmatrix} 2 \\ -1 \\ 7 \end{pmatrix} - \begin{pmatrix} -1 \\ 2 \\ -2 \end{pmatrix} = \begin{pmatrix} 3 \\ -3 \\ 9 \end{pmatrix} = 3 \begin{pmatrix} 1 \\ -1 \\ 3 \end{pmatrix}$ normal vector of Π_2 is $\begin{pmatrix} 1 \\ -1 \\ 3 \end{pmatrix} \times \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} = \begin{pmatrix} 3 \\ 3 \\ 0 \end{pmatrix} = 3 \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$ Therefore the equation of Π_2 is $\mathbf{r} \cdot \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} \cdot \begin{pmatrix} -1 \\ 2 \\ -2 \end{pmatrix} = 1$

(ii)	$l_{AN} : \vec{r} = \begin{pmatrix} -1 \\ 2 \\ -2 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}$ Since N lies on the plane, then we have $\begin{pmatrix} -1+\lambda \\ 2-\lambda \\ -2 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} = 5 \Rightarrow -1+\lambda-2+\lambda=5 \Rightarrow \lambda=4$ Therefore $\overrightarrow{ON} = \begin{pmatrix} 3 \\ -2 \\ -2 \end{pmatrix}$
(iii)	$\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} \times \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ -2 \end{pmatrix} = -2 \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$ Thus the equation of line of intersection is $\vec{r} = \begin{pmatrix} 3 \\ -2 \\ -2 \end{pmatrix} + \lambda \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$
(iv)	$\Pi_3 : \vec{r} \cdot \begin{pmatrix} 4 \\ 2 \\ -1 \end{pmatrix} = D$ Since $\begin{pmatrix} 4 \\ 2 \\ -1 \end{pmatrix} \cdot \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \neq 0$, then the line of intersection is not parallel to Π_3. Therefore the 3 planes meet at a common point.

(v)

Let $\overrightarrow{OC} = \begin{pmatrix} D/4 \\ 0 \\ 0 \end{pmatrix}$ where C lies on the plane Π_3 .

Then we have

$$\left| \overrightarrow{BC} \cdot \begin{pmatrix} 4 \\ 2 \\ -1 \end{pmatrix} \right| \frac{1}{\sqrt{21}} = \sqrt{21}$$

$$\left| \begin{pmatrix} D/4 - 2 \\ 1 \\ -7 \end{pmatrix} \cdot \begin{pmatrix} 4 \\ 2 \\ -1 \end{pmatrix} \right| = 21$$

$$|D + 1| = 21$$

$$D + 1 = 21 \quad \text{or} \quad D + 1 = -21$$

$$D = 20 \quad \text{or} \quad D = -22 \quad (\text{NA})$$

OR

Let N be the foot of perpendicular from B to Π_3 .

$$\overrightarrow{BN} = \pm \sqrt{21} \begin{pmatrix} 4 \\ 2 \\ -1 \end{pmatrix} = \pm \begin{pmatrix} 4 \\ 2 \\ -1 \end{pmatrix}.$$

$$\text{Therefore } \overrightarrow{ON} = \overrightarrow{OB} + \overrightarrow{BN} = \begin{pmatrix} 6 \\ 1 \\ 6 \end{pmatrix} \text{ or } \begin{pmatrix} -2 \\ -3 \\ 8 \end{pmatrix}.$$

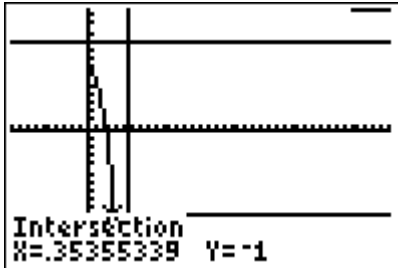
Hence

$$\begin{pmatrix} 6 \\ 1 \\ 6 \end{pmatrix} \cdot \begin{pmatrix} 4 \\ 2 \\ -1 \end{pmatrix} = 21 \quad \text{or} \quad \begin{pmatrix} -2 \\ -3 \\ 8 \end{pmatrix} \cdot \begin{pmatrix} 4 \\ 2 \\ -1 \end{pmatrix} = -22$$

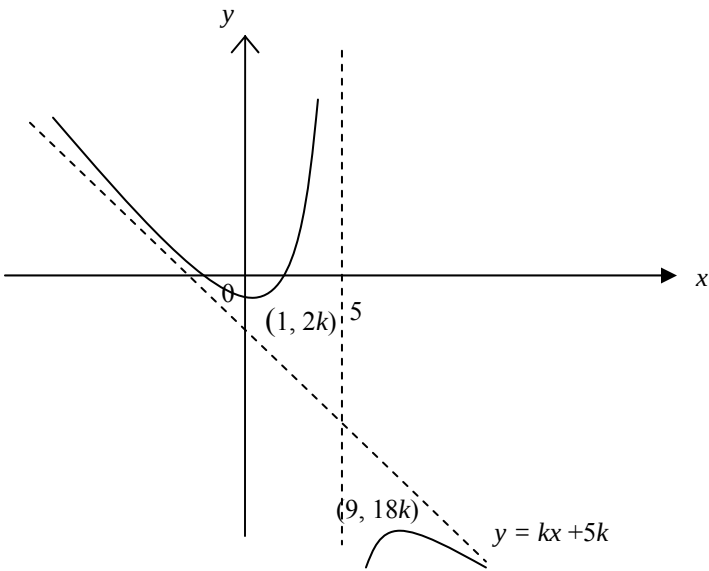
2009 NJC SH2 H2 Mathematics Prelim Paper 1 Solution

11	A sequence A has first term $\ln(2x)$ and the sum of the first n terms is $\frac{n}{2} \ln(2^{n+1} x^2)$ where x is an unknown constant. (i) Show that the n^{th} term of the sequence is $\ln(2^n x)$. Prove that A is an arithmetic progression. [4] The first two terms of sequence A forms two consecutive terms of a geometric progression G respectively. Write down the common ratio of G. [1] (ii) Determine the range of values of x such that G is convergent. [2] (iii) Taking $x = 0.2$, find the first term of G such that its sum to infinity is half of the sum of the first 40 terms of A. [2]
(i)	$ \begin{aligned} S_n - S_{n-1} &= \frac{n}{2} \ln(x^2 2^{n+1}) - \frac{n-1}{2} \ln(x^2 2^n) \\ &= \frac{n}{2} \ln(x^2) + \frac{n}{2} \ln 2^{n+1} - \frac{n-1}{2} \ln(x^2) - \frac{n-1}{2} \ln 2^n \\ &= \frac{1}{2} \ln(x^2) + \frac{n}{2} \ln 2^n + \frac{n}{2} \ln 2 - \frac{n-1}{2} \ln 2^n + \frac{1}{2} \ln 2^n \\ &= \ln(x) + n \ln 2 \\ &= \ln(2^n x) \text{ (shown)} \end{aligned} $ $ \begin{aligned} T_n - T_{n-1} &= \ln(x 2^n) - \ln(x 2^{n-1}) \\ &= \ln x - \ln 2^n - \ln x + \ln 2^{n-1} \\ &= \ln\left(\frac{2^n}{2^{n-1}}\right) \\ &= \ln 2 \end{aligned} $ Since $T_n - T_{n-1} = \text{constant}$, A is an arithmetic progression.
	$ \frac{\ln(4x)}{\ln(2x)} = \frac{\ln(4x)}{\ln(2x)} $ Common ratio: $\frac{\ln(4x)}{\ln(2x)}$

2009 NJC SH2 H2 Mathematics Prelim Paper 1 Solution

(ii)	For G is convergent, $-1 < \frac{\ln(4x)}{\ln(2x)} < 1$  Range of x: $0 < x < 0.354$
(iii)	Let a be the first term in G, Taking $x = 0.2$, $\frac{a}{1 - \frac{\ln 0.8}{\ln 0.4}} = \frac{1}{2} \left(\frac{40}{2} \right) (2 \ln(0.4) + 39 \ln 2)$ $\Rightarrow a = \left(1 - \frac{\ln 0.8}{\ln 0.4} \right) (252.0015858)$ $\therefore a = 190.63184 \approx 191$
12	The curve C is the graph of $f(x) = \frac{k(x^2 - a^2)}{x - 5}$, k and a are real numbers with $a > 0$ and $k < 0$. C has a minimum point at $x = 1$. (i) Show that $a = 3$. [2] Leave your answers in terms of k, (ii) obtain the equations of the asymptotes of C. [3] (iii) determine the coordinates of the turning points of C. [3] (iv) sketch the curve C and the graph of $y = f'(x)$ on two separate diagrams. Label your graphs clearly. [4]

2009 NJC SH2 H2 Mathematics Prelim Paper 1 Solution

(i)	$f(x) = k \left(x + 5 + \frac{25 - a^2}{x - 5} \right)$ $f'(x) = k \left(1 - \frac{25 - a^2}{(x - 5)^2} \right)$ Since stationary point at $x = 1$, therefore $1 - \frac{25 - a^2}{(1 - 5)^2} = 0 \Rightarrow a^2 = 9 \Rightarrow a = 3.$
(ii)	By long division, $f(x) = k \left(x + 5 + \frac{25 - a^2}{x - 5} \right)$ Vertical asymptote: $x = 5$ Oblique asymptote: $y = kx + 5k$
(iii)	For other stationary point, $1 - \frac{25 - 9}{(x - 5)^2} = 0 \Rightarrow x = 9$ Hence the stationary points of C are $(1, 2k)$ and $(9, 18k)$.
(iv)	 The graph shows a function with a vertical asymptote at $x = 5$ and an oblique asymptote $y = kx + 5k$. The curve has a local minimum at $(1, 2k)$ and a local maximum at $(9, 18k)$. The origin is marked with 0.

